



M-26-14 and Elastic

Reference architectures for meeting OMB M-26-14 logging and retention requirements using the Elastic Search AI Platform

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Executive Summary

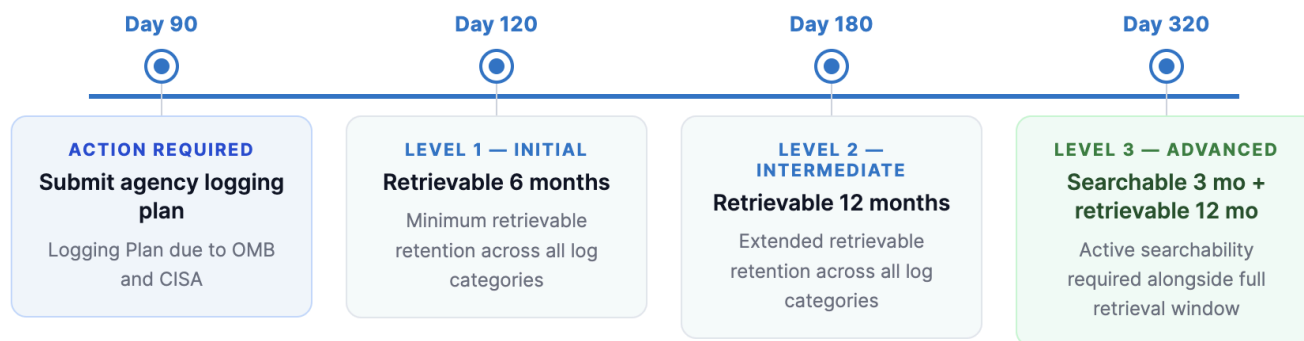
OMB Memorandum M-26-14, issued May 2026, establishes new federal requirements for how agencies collect, retain, and access log data from their information systems. Compliance is not optional — agencies face mandatory maturity milestones beginning within months of the CISA Logging Reference Architecture publication, with full compliance required within a year. Elastic's Search AI Platform is purpose-built for exactly this mission: providing the search, retention, and detection capabilities agencies need to meet M-26-14 requirements from day one, at the scale their environment demands. This document describes three reference architectures — Small, Medium, and Large — designed to match an agency's data volume to the right deployment configuration, and provides guidance for selecting the tier that fits both current obligations and future growth.

The M-26-14 Mandate: What It Requires and When

Issued May 22, 2026, OMB Memorandum M-26-14 — *Ensuring Effective and Efficient Agency Logging and Network Visibility to Defend Against Evolving Cyber Threats* — replaces the prior M-21-31 logging mandate with a risk-based framework built around two core objectives:

 CEM Continuous event monitoring	 THIRF Threat hunting, investigation & forensics
The ability to monitor network activity in real time, detect anomalous behavior as it occurs, and respond promptly. This is your SOC's day-to-day operational capability. 🕒 Searchable minimum 6 months at Level 4 📁 Hot + cold tier data ✔ Elastic delivers full CEM coverage	The ability to investigate and reconstruct network activity after a known or suspected compromise — looking back across months of log history to map how an attacker moved and what they accessed. 🕒 Retrievable minimum 12 months at Level 4 📁 Frozen tier / unmounted snapshots ✔ Elastic delivers full THIRF coverage

M-26-14 sets minimum retention requirements to support both objectives: log data must remain immediately searchable for at least six months, and retrievable for at least twelve months. These are not day-one requirements — they represent the Optimal (Level 4) target that agencies progress toward through a defined maturity model. However, the clock is already running.



The CISA Logging Reference Architecture that starts these compliance clocks is due by August 2026. Agencies that are not already moving are at risk of falling behind before the first deadline arrives.

How Elastic Addresses M-26-14

The Elastic Search AI Platform is a distributed, real-time analytics engine purpose-built to ingest, store, search, and analyze data at enterprise scale. Kibana provides interactive dashboards and visualization, Elastic Agent standardizes data collection across an agency's entire infrastructure footprint — including IT, OT, and IoT — and a library of over 500 integrations accelerates onboarding from virtually any data source. Native machine learning capabilities support real-time anomaly detection, automated alerting, and AI-enhanced threat analysis, directly aligned with M-26-14's emphasis on automated detection and response.

Elastic's tiered storage model — spanning hot, cold, and frozen data tiers — is what makes M-26-14 compliance economically viable at scale. Data is kept immediately searchable in high-performance storage during the active retention window, then moves automatically to progressively lower-cost tiers as it ages. Long-term retention is handled via object storage, where data remains fully retrievable on demand without the cost of keeping it in active search capacity. The result is a cost curve that grows far more slowly than the data it retains.



Elastic's tiered storage model means agencies retain twelve months of log data at a fraction of the cost of keeping it all immediately searchable — without sacrificing the ability to retrieve any of it on demand.

Why Elastic for M-26-14

Elastic is not a general-purpose platform being adapted to meet M-26-14 — it is the platform the requirements were written around. Elastic Cloud Hosted (ECH) is FedRAMP High authorized and

available across commercial and government cloud environments. The Elastic M-26-14 Compliance Pack provides purpose-built detection rules, ML jobs, and compliance dashboards that give agencies an accelerated path to Advanced (Level 3) maturity in weeks, not months. The platform complements existing security investments rather than replacing them, and is fully aligned with the CISA Zero Trust Maturity Model.



AUTHORIZATION

FedRAMP High authorized

Elastic Cloud Hosted is authorized at the FedRAMP High impact level, meeting the most stringent federal cloud security requirements across commercial and government cloud environments.

 Available on commercial & GovCloud



INTEGRATIONS

500+ pre-built integrations

A library of over 500 integrations covers virtually every federal data source — from endpoint telemetry and network logs to cloud platforms, OT systems, and identity providers — out of the box.

 Covers all M-26-14 Appendix B sources



COMPLIANCE

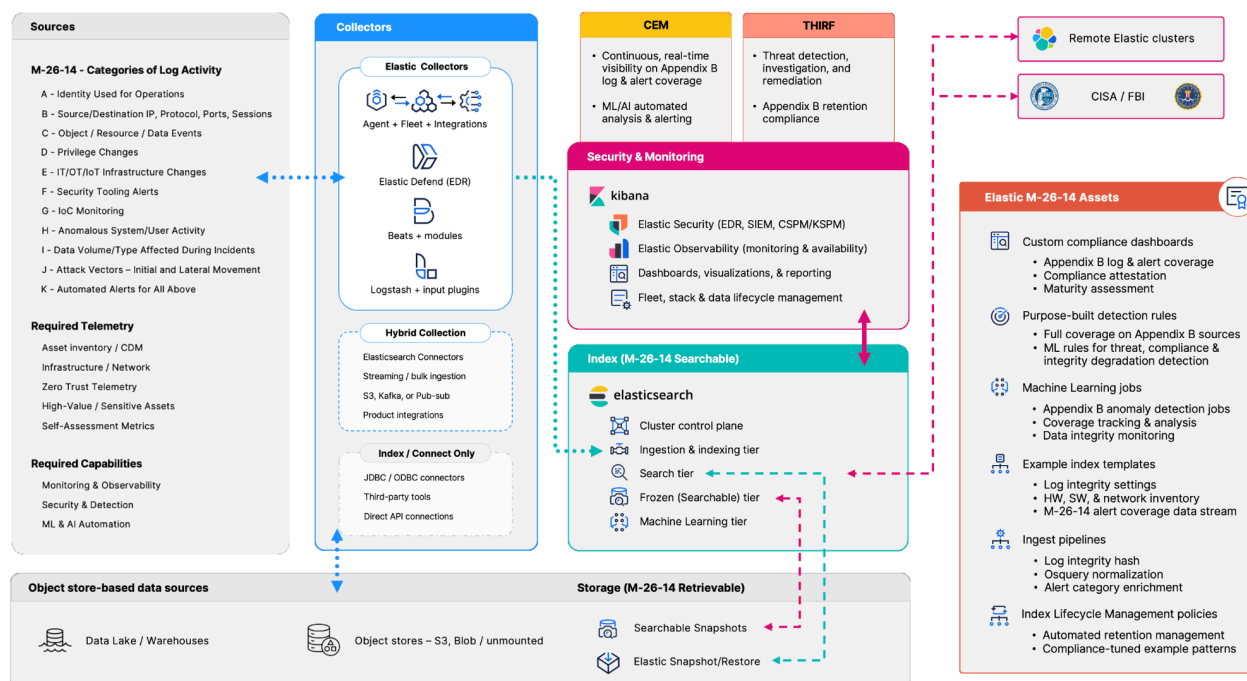
CISA Zero Trust aligned

Elastic's architecture supports the Visibility and Analytics capabilities that underpin all five pillars of the CISA Zero Trust Maturity Model, complementing M-26-14 compliance with a broader security posture.

 Supports all five Zero Trust pillars

Reference Architecture

The diagram below illustrates the full scope of Elastic's M-26-14 coverage — from data collection across every Appendix B log category, through the CEM and THIRF capabilities the mandate requires, to the pre-built compliance assets available in the Elastic M-26-14 Compliance Pack. Every element an agency needs to meet the mandate is represented.



Left side: every M-26-14 Appendix B log category and data source — Elastic collects all of them. **Center:** the CEM and THIRF capabilities your agency needs to demonstrate. **Right side:** the Elastic assets that come pre-built in the Compliance Pack

Elastic deployment options

Elastic is designed for the realities of federal infrastructure — diverse, distributed, and often subject to strict data handling requirements. The platform supports deployment across the full spectrum of federal environments:

OPTION 1 Fully managed cloud	OPTION 2 Self-managed
Elastic manages the underlying infrastructure, upgrades, and scaling. The fastest path to M-26-14 compliance with no operational overhead.	Full platform capabilities on agency-owned or agency-controlled infrastructure. Supports air-gapped and disconnected environments.
AVAILABLE PLATFORMS	AVAILABLE PLATFORMS
 Elastic Cloud Hosted (ECH) Commercial AWS, Azure, and GCP — FedRAMP High authorized	 Elastic Cloud on Kubernetes (ECK) EKS, AKS, GKE, and standard Kubernetes distributions
 ECH on GovCloud FedRAMP Moderate and High accredited government cloud environments	 Self-hosted infrastructure Physical, VM, and hyperconverged hardware
	 Air-gapped enclaves Fully disconnected and isolated network environments

Elastic M-26-14 Compliance Pack

Meeting M-26-14 requirements typically involves standing up detection rules, configuring retention policies, building compliance dashboards, and validating coverage across all Appendix B log categories — work that can take months when built from scratch. The Elastic M-26-14 Compliance Pack eliminates that build time by providing a purpose-built set of assets that address the mandate's requirements out of the box.

The Compliance Pack includes pre-built detection rules covering all M-26-14 Appendix B log activity categories, machine learning jobs for anomaly detection and maturity measurement, index lifecycle management policies pre-configured for Level 3 and Level 4 retention requirements, and Kibana dashboards for compliance attestation, maturity tracking, and Appendix B coverage reporting.

The practical result is that agencies deploying Elastic with the Compliance Pack can achieve Advanced (Level 3) maturity in a matter of weeks — well ahead of the 320-day mandate deadline — and enter a continuous operating state at Optimal (Level 4) shortly thereafter.

Compliance objective	Elastic delivery
 CEM Log collection	A single unified agent collects logs, metrics, and security telemetry across IT, OT, and IoT infrastructure, with over 500 pre-built integrations covering virtually every federal data source.
 CEM Asset inventory	Direct integrations with CDM, HWAM, and SWAM platforms provide the asset visibility required to demonstrate collection coverage across the agency's full inventory.
 THIRF Detection	Purpose-built detection rules covering every M-26-14 Appendix B log activity category, plus approximately 1,800 pre-built Elastic Security rules for broader threat coverage.
 THIRF Anomaly detection	Machine learning jobs purpose-built for M-26-14 automatically detect anomalous activity, track coverage across maturity elements, and generate alerts aligned to mandate requirements.
 THIRF Threat intelligence	Native integration with CISA Automated Indicator Sharing (AIS) and MS-ISAC TAXII feeds, automatically matched against collected log data for real-time indicator detection.
 APPENDIX C Maturity measurement	Dedicated machine learning jobs measure agency progress across each maturity model element, with a Maturity Dashboard providing a real-time view of compliance posture.
 ATTESTATION Compliance attestation	A purpose-built Compliance Attestation Dashboard and exportable coverage matrix give agencies the evidence needed to demonstrate maturity progress to OMB and CISA.



Agencies deploying Elastic with the M-26-14 Compliance Pack can achieve Advanced (Level 3) maturity in weeks — well ahead of the 320-day mandate deadline.

Reference Architecture Tiers

Three reference architectures are defined, each matched to a range of daily data volumes. Customers should select the tier that covers their peak sustained volume with room to grow — sizing to the average rather than the peak is the most common cause of remediation after initial deployment.



TIER 1
Small

Up to 2 TB/day
Up to ~403 TB retained

Suited for a single agency component, departmental security program, or early-stage observability platform.

TYPICAL USE CASES

- ✓ Departmental SOC
- ✓ Application monitoring
- ✓ Dev/test environments
- ✓ FedRAMP High • Multi-AZ • Zero Trust



TIER 2
Medium

2 – 10 TB/day
Up to ~2,015 TB retained

Suited for enterprise environments where multiple teams, programs, or mission areas share a common platform.

TYPICAL USE CASES

- ✓ Enterprise SOC
- ✓ Multi-program observability
- ✓ Compliance-driven data platforms
- ✓ FedRAMP High • Multi-AZ • Zero Trust



TIER 3
Large

10 – 25 TB/day
Up to ~5,037 TB retained

Suited for the most demanding federal deployments where Elastic serves as the backbone of an agency-wide data and detection platform.

TYPICAL USE CASES

- ✓ Agency-wide security programs
- ✓ High-throughput SIEM platforms
- ✓ 10,000+ endpoint environments
- ✓ FedRAMP High • Multi-AZ • Zero Trust

All three tiers are designed to meet M-26-14 retention requirements at Optimal (Level 4) maturity and are available on ECH within the FedRAMP boundary.

Selecting the Right Tier

Use the following question to confirm the appropriate tier for your M-26-14 retention obligations. If you are unsure of your starting tier based on data volume, refer to the Reference Architecture Tiers section above.

What are your M-26-14 retention obligations?

Maturity Level	Retention Requirement	Sizing Guidance
Level 1 — Initial	Retrievable minimum 6 months	A Small cluster with frozen tier configured for unmounted snapshots is likely sufficient
Level 2 — Intermediate	Retrievable minimum 12 months	Small may still be appropriate depending on ingest rate; validate frozen node counts against expected object-stored volume before committing
Level 3 — Advanced	Searchable minimum 3 months + retrievable 12 months	Confirm hot and cold tier sizing supports a 3-month searchable window; customers near the upper bound of Small should consider Medium
Level 4 — Optimal	Searchable minimum 6 months + retrievable 12 months	Validate that hot and cold tiers combined cover the full 6-month searchable window; customers near the upper bound of their current tier should size up to ensure headroom
Beyond M-26-14	Agency records schedule or mission-driven retention beyond 12 months	Frozen tier with extended object storage retention is essential regardless of cluster size; treat extended retention requirements as a strong signal to move to Medium or Large if not already there

Scaling as Your Agency Grows

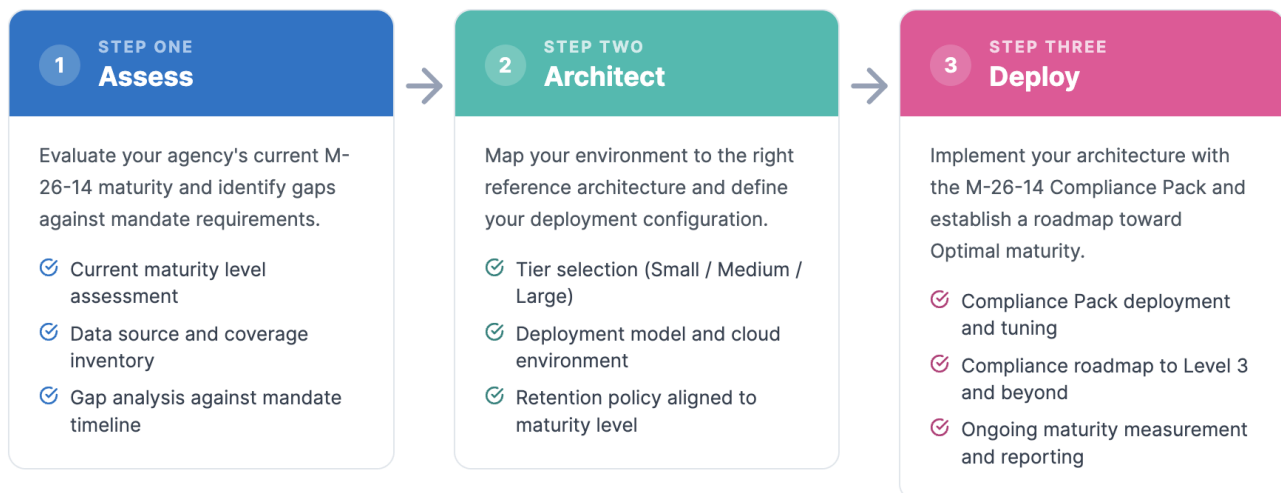
The reference architectures in this document are starting points, not hard limits. As an agency's data volume grows, mission scope expands, or M-26-14 maturity obligations increase, the platform scales to match.

Moving between tiers — Small to Medium, or Medium to Large — is an online operation. Elastic Cloud Hosted manages the underlying infrastructure changes automatically, redistributing data across new capacity without downtime or disruption to active operations. There is no migration, no re-ingestion, and no service window required. Agencies should plan capacity reviews at least quarterly and initiate sizing upgrades proactively; the platform supports growth, but leading it is always preferable to chasing it.

For data volumes above 25 TB/day, or for environments with requirements outside the standard reference model, Elastic's federal solutions team can develop a custom sizing engagement tailored to the agency's specific mission and infrastructure context.

Next Steps

Elastic's federal team is available to help your agency assess its current M-26-14 maturity, map existing data sources and infrastructure to the appropriate reference architecture, and develop a deployment and compliance roadmap aligned to the mandate's timeline.



To start that conversation, contact your Elastic account team or visit elastic.co/federal.